

Time-Domain Analysis of Discrete-Time Systems

Chapter 5

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Outline

- Iterative Solutions to Difference Equations
- The Zero-Input Response
- The Unit Impulse Response
- The Zero-State Response
- Total Response
- System Stability
- Intuitive Insight into System Behavior

Iterative Solutions to Difference Equations

From Chapter 4

$$\left. \frac{d^K}{dt^K} y(t) \right|_{t=nT} \approx \frac{1}{T^K} \sum_{k=0}^K (-1)^k \binom{K}{k} y[n-k].$$

A system can be described by the K^{th} order difference equation.

$$y[n] = -a_1 y[n-1] - \cdots - a_{K-1} y[n-(K-1)] - a_K y[n-K] + \\ b_0 x[n] + b_1 x[n-1] + \cdots + b_{K-1} x[n-(K-1)] + b_K x[n-K]$$

The output $y[n]$ at the n^{th} instant is computed from $2K + 1$ pieces of information. These are the past K output values $y[n-1], y[n-2], \dots, y[n-K]$, the past K input values $x[n-1], x[n-2], \dots, x[n-K]$, and the present input value $x[n]$.

Example for Iterative Solution

Example: Using initial condition $y[-1] = 6$ and causal input $x[n] = \cos(n\pi/2)u[n]$, iteratively solve the first-order system given by $y[n] - 0.5y[n-1] = x[n] - 0.5x[n-1]$.

$$y[n] = 0.5y[n-1] + x[n] - 0.5x[n-1]$$

$$\begin{aligned} y[0] &= 0.5y[-1] + x[0] - 0.5x[-1] \\ &= 0.5(6) + 1 - 0.5(0) = 4. \end{aligned}$$

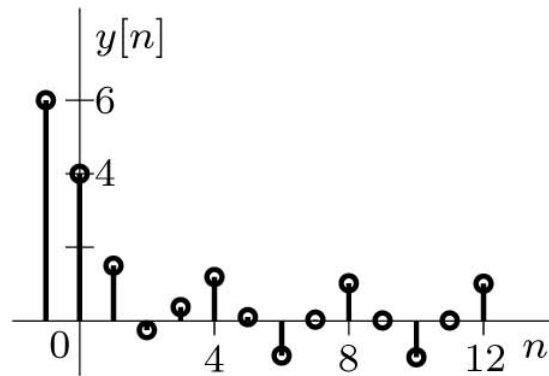
$$y[1] = 0.5(4) + 0 - 0.5(1) = 1.5$$

$$y[2] = 0.5(1.5) - 1 - 0.5(0) = -0.25$$

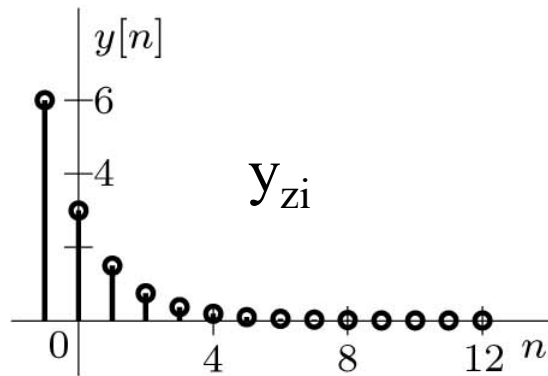
$$y[3] = 0.5(-0.25) + 0 - 0.5(-1) = 0.3750$$

$$y[4] = 0.5(0.3750) + 1 - 0.5(0) = 1.1875$$

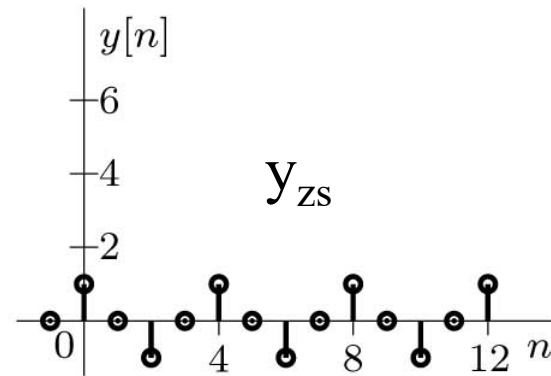
```
n = (-1:12); x = cos(pi*n/2).*(n>=0);  
y = zeros(size(n)); y(n==-1) = 6;  
for ind = find(n>=0),  
    y(ind) = 0.5*y(ind-1)+x(ind)-0.5*x(ind-1);  
end  
stem(n,y)
```



(a)



(b)



(c)

Recursive and Nonrecursive Equation

$$y[n] = -a_1y[n-1] - \cdots - a_{K-1}y[n-(K-1)] - a_Ky[n-K] + b_0x[n] + b_1x[n-1] + \cdots + b_{K-1}x[n-(K-1)] + b_Kx[n-K] \quad \text{IIR System}$$

When $a_k = 0$ for $k = (1, \dots, K)$ then the equation is nonrecursive

$$y[n] = b_0x[n] + b_1x[n-1] + \cdots + b_Kx[n-K] \quad \text{FIR System}$$

To calculate the output $y[n]$ we don't need past values of the outputs so there is no need to calculate the output recursively.

Example

A digital differentiator is represented as

$$y[n] = \frac{1}{T}(x[n] - x[n-1])$$

Determine the response of this system to **(a)** $x[n] = u[n]$ and **(b)** $x[n] = nT u[n]$.

Recursive versus Nonrecursive Implementations

Recursive require less operation to find the output in comparison to nonrecursive.

Example

Consider a five-point sum, where the current output is simply a sum of the five most recent input values. Express this system using both nonrecursive and recursive forms, and compare the two implementations.

Nonrecursive form requires 4 additions:

$$y[n] = x[n] + x[n - 1] + x[n - 2] + x[n - 3] + x[n - 4].$$

4 operations + 4 memories

Recursive form requires 2 additions:

$$y[n] = x[n] + y[n - 1] - x[n - 5]$$

2 operations + 2 memories

5.2

Operator Notation

Operator Notation

$E\{ \}$ notation stands for time shift

$$E\{x[n]\} \equiv x[n + 1]$$

$$E^2\{x[n]\} \equiv x[n + 2]$$

$$E^K\{x[n]\} \equiv x[n + K]$$

$$E^{-1}\{x[n]\} \equiv x[n - 1]$$

$$E^{-2}\{x[n]\} \equiv x[n - 2]$$

$$E^{-K}\{x[n]\} \equiv x[n - K]$$

Example: Express the difference equation using the operator notation

$$y[n] - 2y[n - 1] - 7y[n - 2] = x[n] + 3x[n - 1]$$

$$y[n] - 2E^{-1}\{y[n]\} - 7E^{-2}\{y[n]\} = x[n] + 3E^{-1}\{x[n]\}$$

$$(1 - 2E^{-1} - 7E^{-2})\{y[n]\} = (1 + 3E^{-1})\{x[n]\}$$

Multiply by E^2 to convert it to advance form

$$(E^2 - 2E - 7)\{y[n]\} = (E^2 + 3E)\{x[n]\} \quad \longrightarrow \quad A(E)\{y[n]\} = B(E)\{x[n]\}$$

5.3

The Zero-Input Response

The Zero-Input Response

The zero-input response $y[n]$ is the system response to the internal conditions (initial conditions) when the input is zero

$$y[n + K] + a_1 y[n + K - 1] + \cdots + a_{K-1} y[n + 1] + a_K y[n] = 0$$

$$(E^K + a_1 E^{K-1} + \cdots + a_{K-1} E + a_K) \{y[n]\} = 0 \quad \longrightarrow \quad \text{Eq. (1)}$$

The only function $y[n]$ that can be summed with its shifted versions to zero for all values of n is the exponential function γ^n .

$$y[n] = c\gamma^n \quad E\{y[n]\} = y[n+1] = c\gamma^{n+1} = \gamma^1 c\gamma^n \quad E^2\{y[n]\} = y[n+2] = \gamma^2 c\gamma^n$$

$$c(\gamma^K + a_1 \gamma^{K-1} + \cdots + a_{K-1} \gamma + a_K) \gamma^n = 0$$

$$(\gamma^K + a_1 \gamma^{K-1} + \cdots + a_{K-1} \gamma + a_K) = A(\gamma) = 0 \quad (\gamma - \gamma_1)(\gamma - \gamma_2) \cdots (\gamma - \gamma_K) = A(\gamma) = 0$$

The Zero-Input Response

The total zero-input response is a combination of all possible solutions.

$$y[n] = c_1\gamma_1^n + c_2\gamma_2^n + \cdots + c_K\gamma_K^n$$

$A(\gamma) = 0$ is the characteristic equation

$A(\gamma)$ is the characteristic polynomial

$\gamma_1, \gamma_2, \dots, \gamma_K$ are the characteristic roots, eigenvalues

γ_k^n is the characteristic mode

Example

Using initial conditions $y[-1] = 0$ and $y[-2] = 25/4$, determine the zero-input response of an LTID system described by the difference equation

$$y[n + 2] - 0.6y[n + 1] - 0.16y[n] = 5x[n + 2].$$

Solution

$$\left. \begin{aligned} (-0.2)^{-1}c_1 + (0.8)^{-1}c_2 &= 0 \\ (-0.2)^{-2}c_1 + (0.8)^{-2}c_2 &= \frac{25}{4} \end{aligned} \right\} \Rightarrow \begin{bmatrix} -5 & \frac{5}{4} \\ 25 & \frac{25}{16} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{25}{4} \end{bmatrix}$$

$$y[n] = 0.2(-0.2)^n + 0.8(0.8)^n$$

$$>> \mathbf{c} = \text{inv}([-5 \ 5/4; 25 \ 25/16]) * [0; 25/4]$$

Repeated Roots

$$A(\gamma) = (\gamma - \gamma_1)^r (\gamma - \gamma_{r+1}) (\gamma - \gamma_{r+2}) \cdots (\gamma - \gamma_K)$$

$$y[n] = (c_1 + c_2 n + c_3 n^2 + \cdots + c_r n^{r-1}) \gamma_1^n + c_{r+1} \gamma_{r+1}^n + c_{r+2} \gamma_{r+2}^n + \cdots + c_K \gamma_K^n$$

Example: Using initial conditions $y[-1] = -1/3$ and $y[-2] = -2/9$, determine the zero-input response of an LTID system described by the difference equation

$$y[n+2] + 6y[n+1] + 9y[n] = 2x[n+2] + 6x[n+1].$$

Solution

$$\begin{aligned} y[n] &= (c_1 + c_2 n)(-3)^n \\ y[n] &= (4 + 3n)(-3)^n \end{aligned} \quad \begin{bmatrix} (-3)^{-1} & (-1)(-3)^{-1} \\ (-3)^{-2} & (-2)(-3)^{-2} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} \\ -\frac{2}{9} \end{bmatrix}$$

$$\gg c = \text{inv}([(-3)^{-1} \quad (-1)*(-3)^{-1}; (-3)^{-2} \quad (-2)*(-3)^{-2}]) * [-1/3; -2/9]$$

Complex Roots

If one of the characteristic root is complex, then a second root will be its conjugate.

$$\gamma = |\gamma|e^{j\beta} \quad \text{and} \quad \gamma^* = |\gamma|e^{-j\beta}$$

$$y[n] = c_1\gamma^n + c_2(\gamma^*)^n$$

$$= c_1|\gamma|^n e^{j\beta n} + c_2|\gamma|^n e^{-j\beta n}$$

$$c_1 = \frac{c}{2}e^{j\theta} \quad \text{and} \quad c_2 = \frac{c}{2}e^{-j\theta}$$

$$y[n] = \frac{c}{2}|\gamma|^n \left(e^{j(\beta n + \theta)} + e^{-j(\beta n + \theta)} \right)$$

$$= c|\gamma|^n \cos(\beta n + \theta)$$

Example

Using initial conditions $y[-1] = 2$ and $y[-2] = 1$, determine and plot the zero-input response of an LTID system described by the difference equation

$$(E^2 - 1.5588 E + 0.81)\{y[n]\} = (E + 3)\{x[n]\}.$$

Solution

$$\gamma^2 - 1.5588\gamma + 0.81 = 0$$

```
>> gamma = roots([1 -1.5588 0.81])  
gamma = 0.7794+0.4500i 0.7794-0.4500i
```

$$\gamma_1 = 0.78 + j0.45 \text{ and } \gamma_2 = 0.78 - j0.45 = \gamma_1^*$$

$$y[n] = c_1 \underbrace{(0.78 + j0.45)^n}_{\gamma_1} + c_2 \underbrace{(0.78 - j0.45)^n}_{\gamma_2} \quad \begin{bmatrix} \gamma_1^{-1} & \gamma_2^{-1} \\ \gamma_1^{-2} & \gamma_2^{-2} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} y[-1] \\ y[-2] \end{bmatrix}$$

```
c = inv([gamma(1)^(-1) gamma(2)^(-1); gamma(1)^(-2) gamma(2)^(-2)]) * ([2; 1])  
c = 1.1538-0.1984i 1.1538+0.1984i
```

$$y[n] = (1.154 - j0.198)(0.78 + j0.45)^n + (1.154 + j0.198)(0.78 - j0.45)^n$$

Example-Real form solution of Complex Roots

To express the output in a real form $y[n] = c|\gamma|^n \cos(\beta n + \theta)$

$$y[n] = (1.154 - j0.198)(0.78 + j0.45)^n + (1.154 + j0.198)(0.78 - j0.45)^n$$

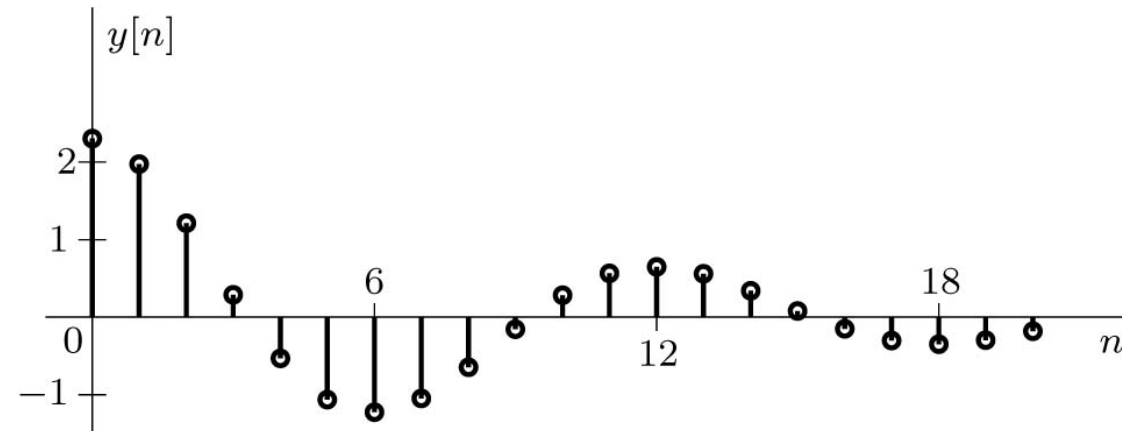
then find the magnitude and angle of the complex roots and constants.

$$\gamma_1 = |\gamma|e^{j\beta} \quad c_1 = \frac{c}{2}e^{j\theta}$$

$$\gamma_1 = 0.78 + j0.45 = 0.9e^{j\pi/6} \quad \text{so} \quad |\gamma| = 0.9 \text{ and } \beta = \pi/6$$

$$c_1 = 1.154 - j0.1984 = 1.171e^{-j0.17} \quad \text{so} \quad c = 2 c_1 = 2.342 \text{ and } \theta = -0.17 \text{ rad}$$

$$y[n] = 2.34(0.9)^n \cos\left(\frac{\pi}{6}n - 0.17\right)$$



Insights into the Zero-Input Behavior of a System

- Combination of the characteristic mode
- Due to energy stored at the system due to previous inputs.
- Resonance Response

5.4

The Unit Impulse Response

The Unit Impulse Response $h[n]$

The K th-order system is described by $A(E) \{y[n]\} = B(E) \{x[n]\}$

The unit impulse response $h[n]$ is the system output for input $\delta[n]$ with all initial conditions are set to zeros (no energy stored in the system).

$$A(E) \{h[n]\} = B(E) \{\delta[n]\} \quad h[-1] = h[-2] = \dots = h[-K] = 0$$

To determine $h[n]$, the above equation can be solved iteratively or in a closed form.

Example 5.8: Iteratively find and then plot the unit impulse response $h[n]$ of a system described by the equation

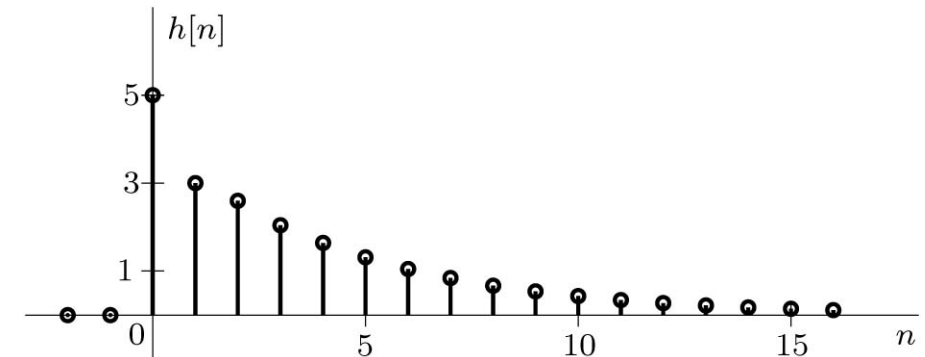
$$y[n] - 0.6y[n-1] - 0.16y[n-2] = 5x[n].$$

Solution

$$h[0] - 0.6(0) - 0.16(0) = 5(1)$$

$$h[0] = 5$$

$$h[1] = 3, \quad h[2] = 2.6$$



```
n = (-2:16); delta = (n==0); h = zeros(size(n));  
for ind = find(n>=0),  
    h(ind) = 0.6*h(ind-1)+0.16*h(ind-2)+5*delta(ind);  
end  
stem(n,h)
```

The Unit Impulse Response-Closed Form

The unit impulse response will consist of response at $n = 0$ due to the impulse and combinations of the characteristic modes for $n > 0$ since the impulse input does not exist for $n \neq 0$.

$$y[n] + a_1y[n-1] + a_2y[n-2] + \dots + a_Ky[n-K] = b_0x[n] + b_1x[n-1] + \dots + b_Lx[n-L]$$

First Case: $L < K$

$$h[n] = c_1(\gamma_1)^n + c_2(\gamma_2)^n + \dots + c_K(\gamma_K)^n$$

Second Case: $L = K$

$$h[n] = (b_K/a_K)\delta[n] + c_1(\gamma_1)^n + c_2(\gamma_2)^n + \dots + c_K(\gamma_K)^n$$

Third Case: R Zero Characteristic Roots

$$h[n] = A_0\delta[n] + A_1\delta[n-1] + \dots + A_R\delta[n-R] + c_1(\gamma_1)^n + c_2(\gamma_2)^n + \dots + c_{K-R}(\gamma_{K-R})^n$$

The constants $c_1, c_2, \dots$ and $A_0, A_1, \dots$ can be found by using $h[0], h[1], \dots, h[K]$ as initial conditions and then solve the polynomial equations. $h[0], h[1], \dots, h[K]$ can be found by iterative calculation.

Example-Finding $h[n]$ in closed Form

Determine a closed-form representation of the unit impulse response $h[n]$ for the system specified by the difference equation

$$y[n] - 0.6y[n - 1] - 0.16y[n - 2] = 5x[n]$$

Solution

$$(E^2 - 0.6E - 0.16) \{y[n]\} = 5E^2 \{x[n]\}$$

$$\gamma^2 - 0.6\gamma - 0.16 = (\gamma + 0.2)(\gamma - 0.8)$$

$$y_c[n] = c_1(-0.2)^n + c_2(0.8)^n$$

$$h[n] = (c_1(-0.2)^n + c_2(0.8)^n)u[n]$$

By iterative calculation with $h[n] = 0$ for all $n < 0$ we can find $h[0] = 5$ and $h[1] = 3$. Use these values to solve for the constants c_1 and c_2 in the above equation. ($c_1 = 1$ and $c_2 = 4$).

$$h[n] = [(-0.2)^n + 4(0.8)^n] u[n]$$

Example-Finding $h[n]$ with Zero Characteristic Roots

Determine the impulse response $h[n]$ of a system described by the equation $(E^3 + E^2) \{y[n]\} = x[n]$.

Solution

$$\gamma^2(\gamma+1) = 0$$

$$h[n] = A_0\delta[n] + A_1\delta[n-1] + A_2\delta[n-2] + c_1(-1)^n u[n].$$

$$\begin{bmatrix} 1 & 0 & 0 & (-1)^0 \\ 0 & 1 & 0 & (-1)^1 \\ 0 & 0 & 1 & (-1)^2 \\ 0 & 0 & 0 & (-1)^3 \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ A_2 \\ c_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

To solve for A_0, A_1, A_2, c_1 we need to solve the above equation iteratively to obtain values for $h[0], h[1], h[2]$, and $h[3]$.

```
n = (-3:3); delta = (n==0); h =  
zeros(size(n));  
for ind = find(n>=0),  
h(ind) = -h(ind-1)+delta(ind-3);  
end  
h(n>=0)  
ans = 0 0 0 1
```

```
coef = inv([1 0 0 1;0 1 0 -1;0 0 1 1;0 0 0 -1])*([0;0;0;1])  
coef = 1 -1 1 -1
```

$$h[n] = \delta[n] - \delta[n-1] + \delta[n-2] - (-1)^n u[n].$$

Impulse Response of Nonrecursive System

$$y[n] = b_0x[n] + b_1x[n - 1] + \cdots + b_Kx[n - K]$$

Let $x[n] = \delta[n]$ and $y[n] = h[n]$

$$h[n] = b_0\delta[n] + b_1\delta[n - 1] + \cdots + b_K\delta[n - K] = \sum_{k=0}^K b_k\delta[n - k]$$

A nonrecursive system has no natural modes.

5.5

The Zero-State Response

The Zero-State Response

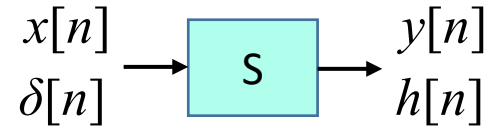
- The zero-state response $y[n]$ is the system response to an input $x[n]$ when the system is in zero state (all initial conditions are zeros).
- Any signal can be expressed as a linear sum of impulses

$$x[n] = \cdots + x[-2] \delta[n+2] + x[-1] \delta[n+1] + x[0] \delta[n] + x[1] \delta[n-1] + x[2] \delta[n-2] + \cdots$$

$$x[n] = \sum_{m=-\infty}^{\infty} x[m] \delta[n - m]$$

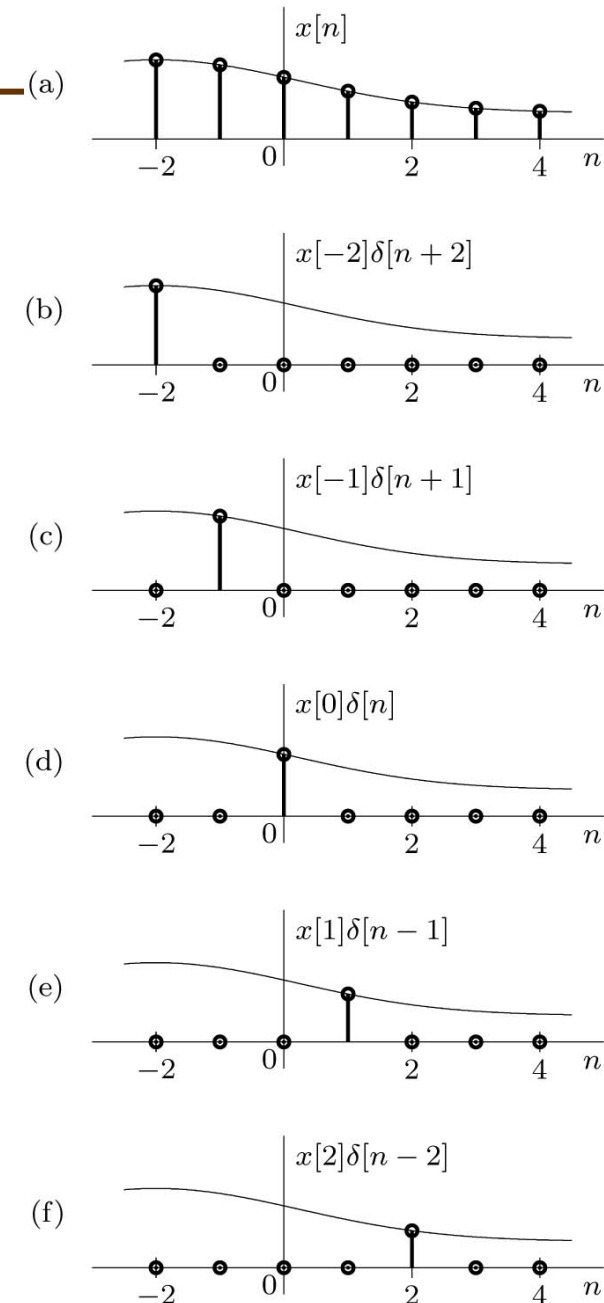
$$y[n] = \sum_{m=-\infty}^{\infty} x[m] \delta[n - m] * h[n]$$

$$y[n] = \sum_{m=-\infty}^{\infty} x[m] h[n - m] \quad \longrightarrow \quad y[n] = x[n] * h[n]$$



$$\delta[n - m] \implies h[n - m]$$

Convolution



The Zero-State Response

For a causal input and causal system, the input output relationship is

$$y[n] = \sum_{m=0}^n x[m]h[n-m]$$

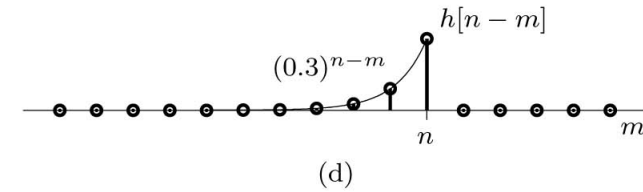
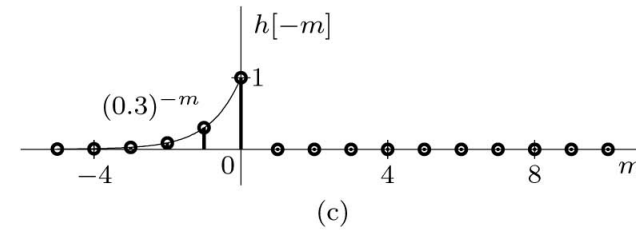
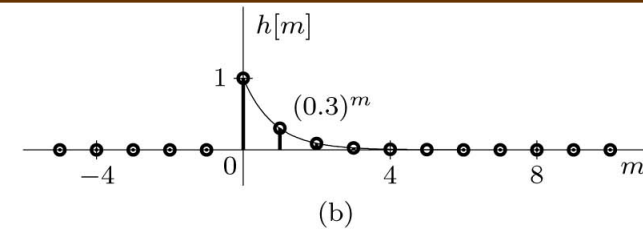
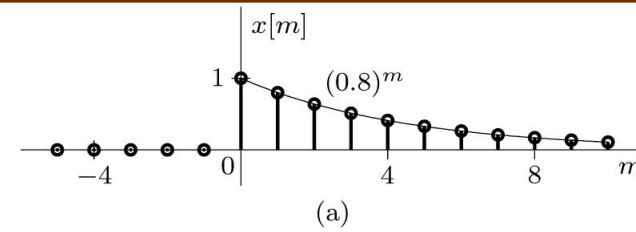
1.	$\sum_{m=p}^n r^m = \frac{r^p - r^{n+1}}{1-r}$	$r \neq 1$
2.	$\sum_{m=0}^n m = \frac{n(n+1)}{2}$	
3.	$\sum_{m=0}^n m^2 = \frac{n(n+1)(2n+1)}{6}$	
4.	$\sum_{m=0}^n m r^m = \frac{r + [n(r-1) - 1]r^{n+1}}{(r-1)^2}$	$r \neq 1$
5.	$\sum_{m=0}^n m^2 r^m = \frac{r[(1+r)(1-r^n) - 2n(1-r)r^n - n^2(1-r)^2 r^n]}{(r-1)^3}$	$r \neq 1$

Example: Analytically determine the DT convolution for $x[n] = (0.8)^n u[n]$ and $h[n] = (0.3)^n u[n]$.

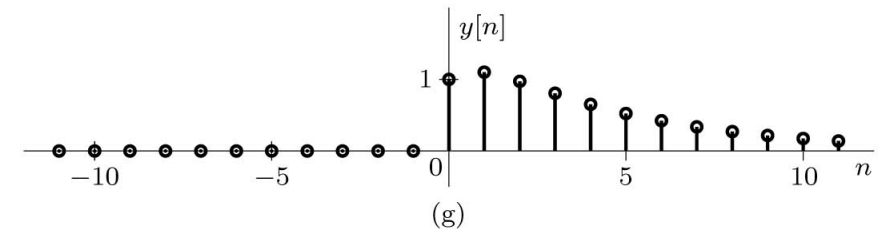
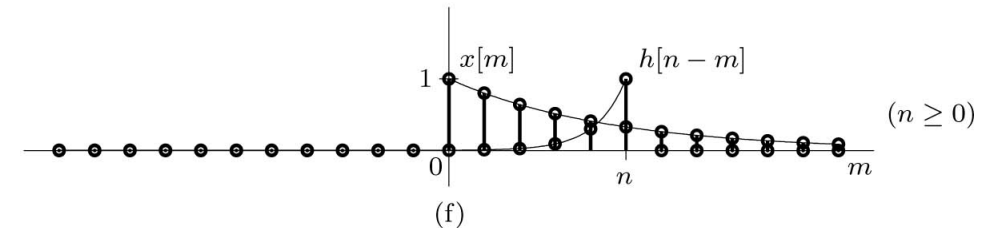
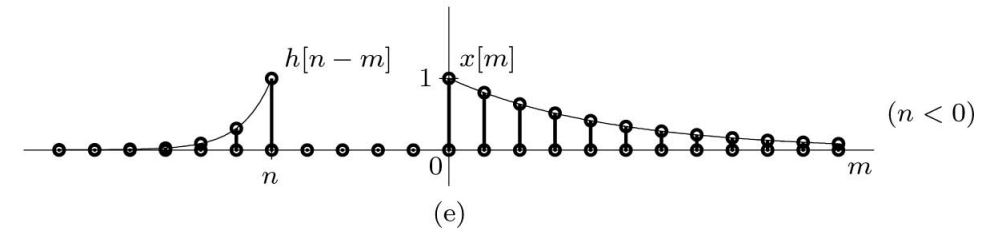
Answer: $y[n] = 2 [(0.8)^{n+1} - (0.3)^{n+1}] u[n]$

Graphical Procedure for the Convolution Sum

$$y[n] = x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m]$$



Example: Using the graphical approach, show the DT convolution $y[n] = x[n] * h[n]$
For $x[n] = (0.8)^n u[n]$ and $h[n] = (0.3)^n u[n]$.



Convolution Sum Properties

Commutative Property: $x[n] * h[n] = h[n] * x[n]$

Distributive Property: $x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$

Associative Property: $x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$

Shifting Property: if $x[n] * h[n] = y[n]$ then $x[n-m] * h[n-p] = y[n-m-p]$

Convolution with an impulse: $x[n] * \delta[n] = x[n]$

Width and Length Property: if $x[n]$ has width W_x (length L_x) and $h[n]$ has width W_h (length L_h) then the width of the convolution $y[n] = x[n] * h[n]$ is $W_y = W_x + W_h$ (length $L_y = L_x + L_h - 1$)

	$x[n]$	$h[n]$	$x[n] * h[n]$
1.	$x[n]$	$\delta[n - k]$	$x[n - k]$
2.	$u[n]$	$u[n]$	$(n + 1)u[n]$
3.	$u[n]$	$n u[n]$	$\frac{n(n+1)}{2} u[n]$
4.	$nu[n]$	$nu[n]$	$\frac{n(n-1)(n+1)}{6} u[n]$
5.	$u[n]$	$\gamma^n u[n]$	$\left(\frac{1-\gamma^{n+1}}{1-\gamma} \right) u[n]$
6.	$nu[n]$	$\gamma^n u[n]$	$\left(\frac{\gamma(\gamma^n - 1) + n(1-\gamma)}{(1-\gamma)^2} \right) u[n]$
7.	$\gamma^n u[n]$	$\gamma^n u[n]$	$(n + 1)\gamma^n u[n]$
8.	$\gamma_1^n u[n]$	$\gamma_2^n u[n]$	$\left(\frac{\gamma_1^{n+1} - \gamma_2^{n+1}}{\gamma_1 - \gamma_2} \right) u[n] \quad \gamma_1 \neq \gamma_2$
9.	$\gamma_1^n u[n]$	$n\gamma_2^n u[n]$	$\frac{\gamma_1 \gamma_2}{(\gamma_1 - \gamma_2)^2} \left(\gamma_1^n + \frac{\gamma_2 - \gamma_1}{\gamma_1} n \gamma_2^n - \gamma_2^n \right) u[n] \quad \gamma_1 \neq \gamma_2$
10.	$\gamma_1^n u[n]$	$\gamma_2^n u[-n - 1]$	$\frac{\gamma_1}{\gamma_2 - \gamma_1} \gamma_1^n u[n] + \frac{\gamma_2}{\gamma_2 - \gamma_1} \gamma_2^n u[-n - 1] \quad \gamma_2 > \gamma_1 $
11.	$ \gamma_1 ^n u[n]$	$ \gamma_2 ^n \cos(\beta n + \theta) u[n]$	$\frac{1}{R} \left[\gamma_2 ^{n+1} \cos[\beta(n+1) + \theta - \phi] - \gamma_1 ^{n+1} \cos(\theta - \phi) \right] u[n]$ $R = \sqrt{ \gamma_1 ^2 - 2 \gamma_1 \gamma_2 \cos(\beta) + \gamma_2 ^2}$ $\phi = \tan^{-1} \left[\frac{ \gamma_2 \sin(\beta)}{ \gamma_2 \cos(\beta) - \gamma_1 } \right]$

Example

Use the convolution properties and the table to find the zero-state response $y[n]$ for the input $x[n] = 4^{-n}u[n]$ of an LTID system described by the equation

$$y[n + 2] - 0.6y[n + 1] - 0.16y[n] = 5x[n + 2];$$

Answer

$$h[n] = [(-0.2)^n + 4(0.8)^n] u[n].$$

$$y[n] = [-1.26(4)^{-n} + 0.444(-0.2)^n + 5.82(0.8)^n]u[n]$$

What if the system is modified as follow

$$y[n + 2] - 0.6y[n + 1] - 0.16y[n] = 5x[n + 1];$$

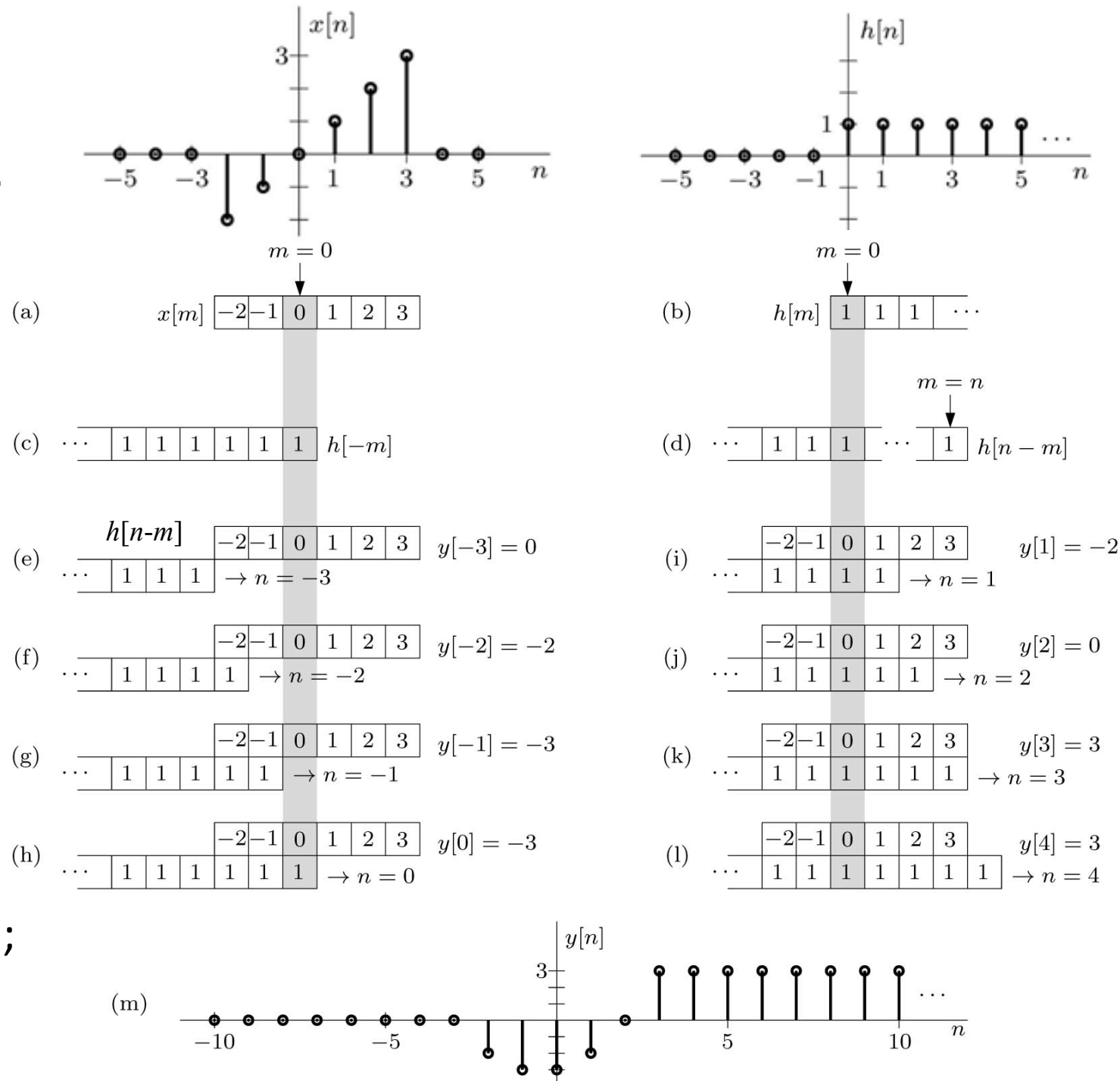
Graphical Procedure: Sliding-Tape Method

$$y[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m]$$

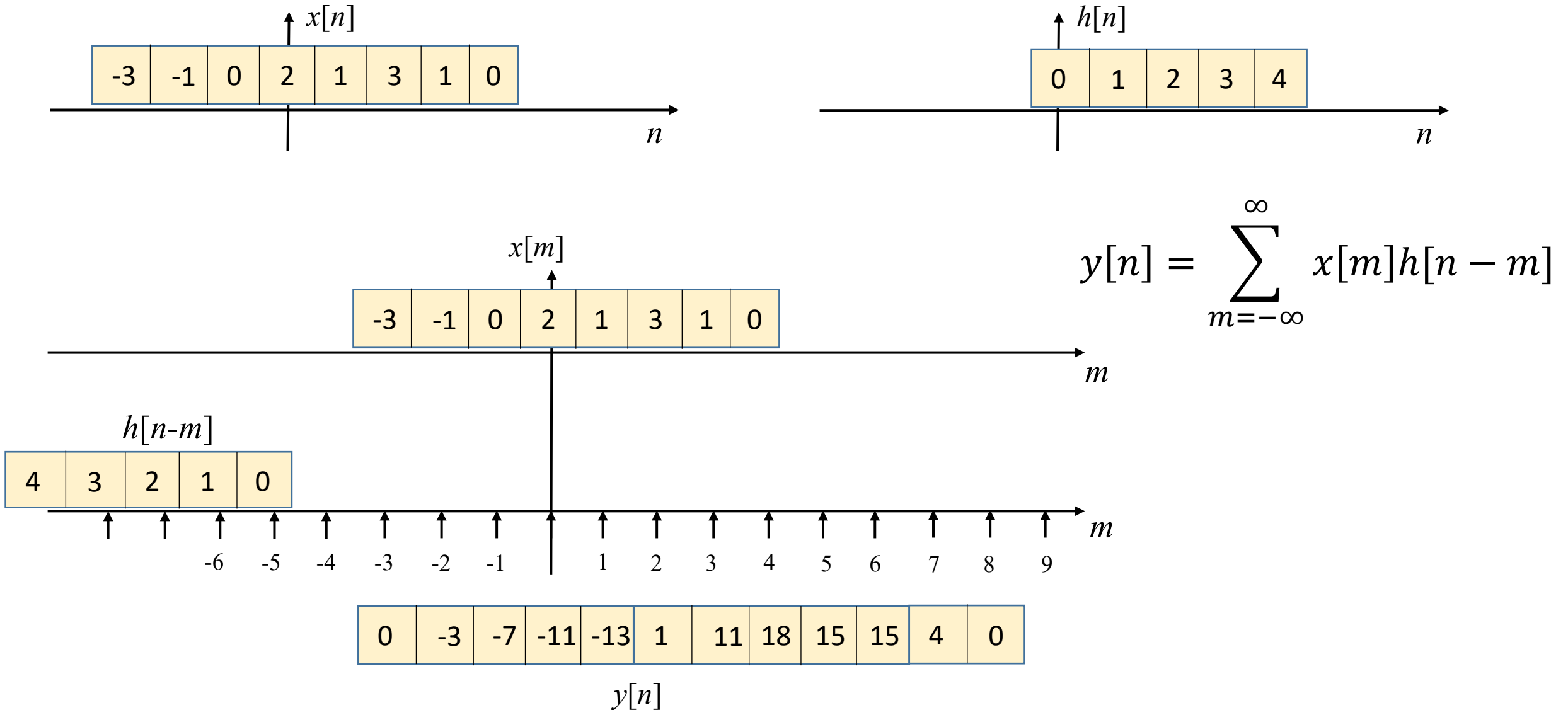
Example: Using the sliding-tape approach, determine the DT convolution $y[n] = x[n] * h[n]$ using the signals $x[n] = n(u[n+2] - u[n-4])$ and $h[n] = u[n]$

Matlab Code

```
x = [-2,-1,0,1,2,3];      h = ones(1,10);
n = (-2:-2+length(x)-1+length(h)-1);
stem(n,conv(x,h));
```

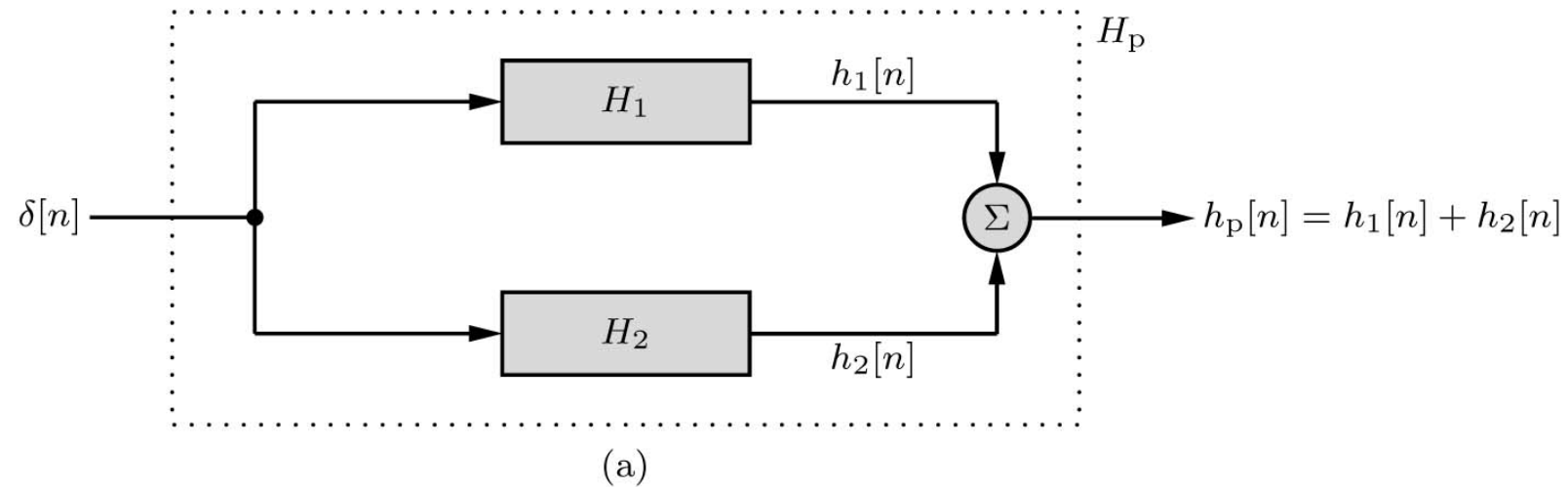


Graphical Procedure: Sliding-Tape Method

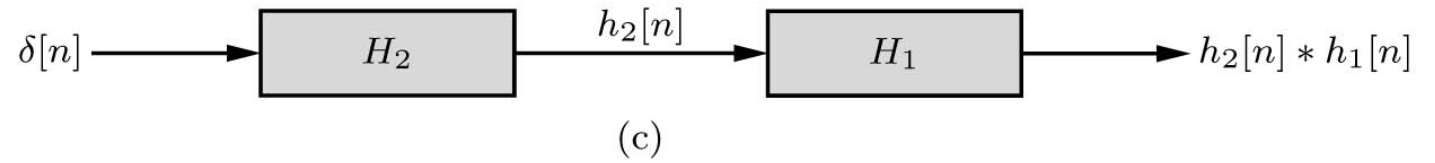
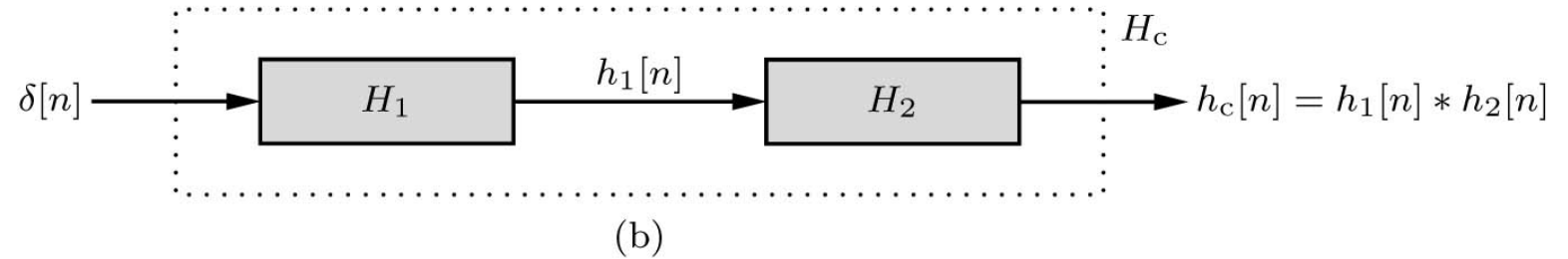


Interconnected Systems

Parallel Connection



Cascade Connection



Example: Show that the inverse of a discrete-time accumulator (CT integrator) is a first-order backward difference system (CT differentiator). Hint: $h[n] * h_i[n] = \delta[n]$.

LTID System Response to an Everlasting Exponential $x[n] = Z^n$

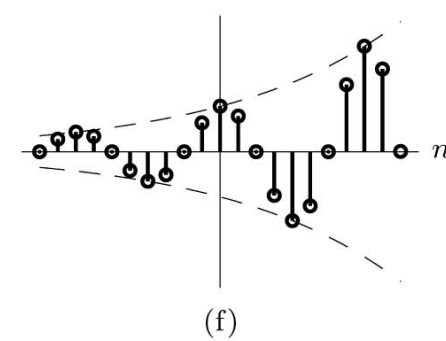
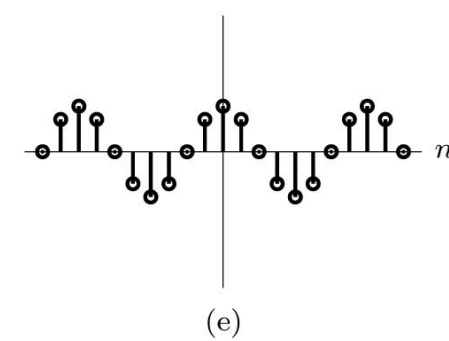
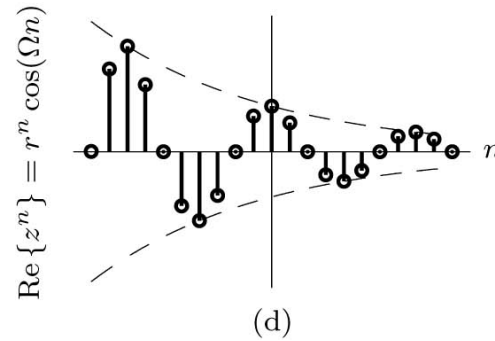
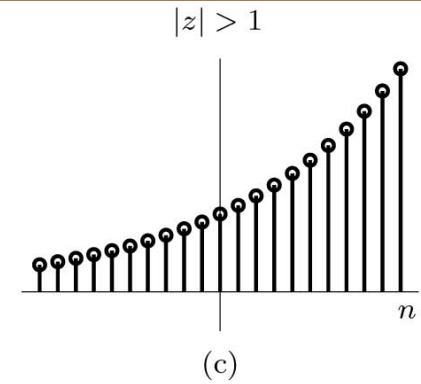
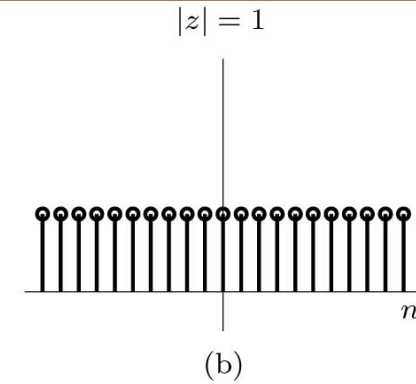
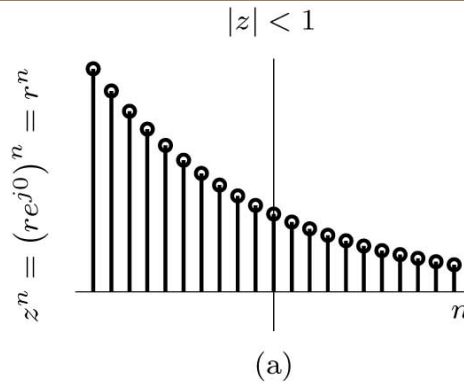
$$y[n] = h[n] * z^n = \sum_{m=-\infty}^{\infty} h[m]z^{n-m}$$

$$= z^n \sum_{m=-\infty}^{\infty} h[m]z^{-m}$$

Let $H(z) = \left. \frac{y[n]}{x[n]} \right|_{x[n]=z^n}$

$$H(z) = \sum_{m=-\infty}^{\infty} h[m]z^{-m}$$

Then $y[n] = H(z)z^n = H(z)x[n]|_{x[n]=z^n}$



$$x[n] = Z^n = (re^{j\Omega})^n$$

For a given z , $H(z)$ is a constant (could be real or complex).

$H(z)$ is the transfer function of the system. It is the system response to different values of z .

The output converges if $H(z)$ converge, and $H(z)$ converges if $|z| < 1$ and the system is stable.

Systems Transfer Function and Frequency Response

For a special case $z = e^{j\Omega n}$

then $y[n] = H(z)z^n = H(z)x[n] \big|_{x[n]=z^n}$

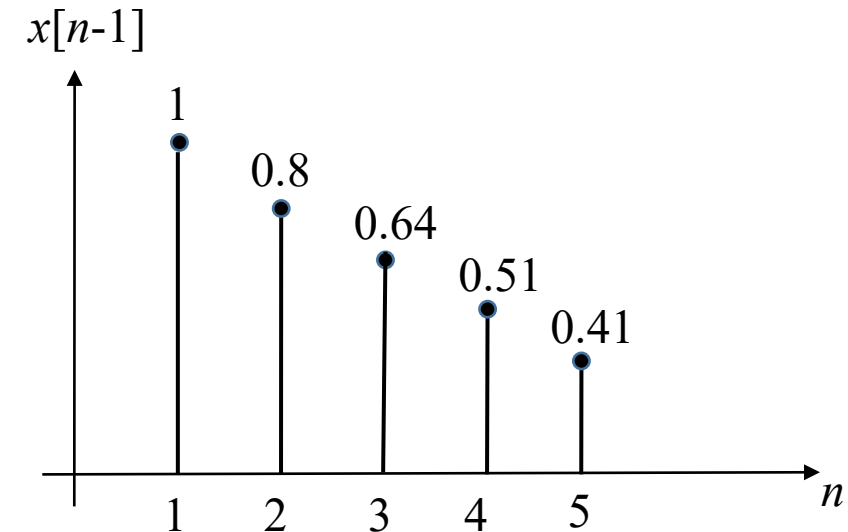
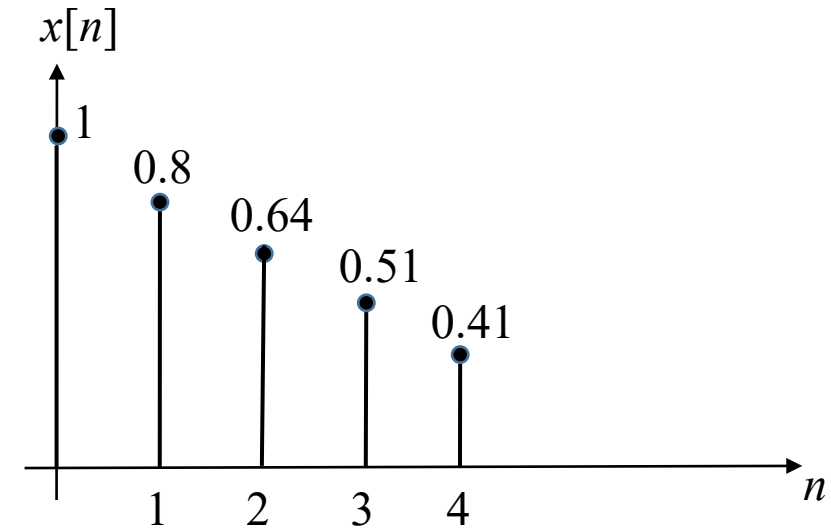
becomes $y[n] = H(e^{j\Omega})e^{j\Omega n} = H(e^{j\Omega})x[n] \big|_{x[n]=e^{j\Omega n}}$

Where $H(e^{j\Omega}) = \sum_{m=-\infty}^{\infty} h[m]e^{-j\Omega m}$ is the frequency response of the system.

Example

Determine the transfer function and frequency response of the following systems:

- a)** digital differentiator given by $y[n+1] = 1/T (x[n+1] - x[n])$
Find the output $y[n]$ for the input $x[n] = (0.8)^n$
- b)** a unit-delay system given by $y[n] = x[n-1]$



5.6

Total Response

Total Response

$$\underbrace{y[n]}_{\text{total response}} = \underbrace{\sum_{k=1}^K c_k \gamma_k^n}_{\text{zero-input response } y_{zir}[n]} + \underbrace{x[n] * h[n]}_{\text{zero-state response } y_{zsr}[n]}$$

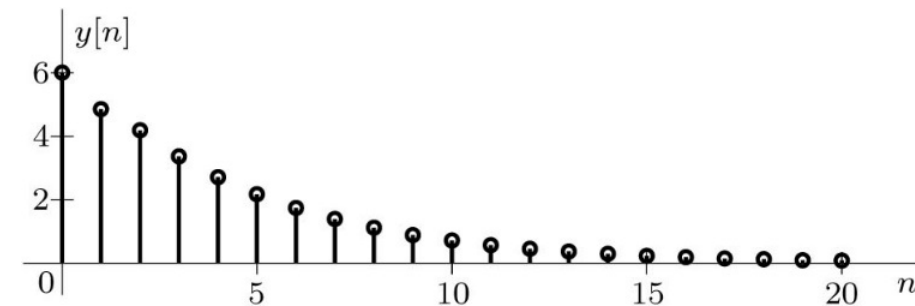
Example: Using the initial conditions $y[-1] = 0$ and $y[-2] = 25/4$ and the input $x[n] = 4^{-n}u[n]$, determine and sketch the total response $y[n]$ of an LTID system described by the equation

$$y[n+2] - 0.6y[n+1] - 0.16y[n] = 5x[n+2].$$

Solution

$$y[n] = \underbrace{[-1.26(4)^{-n} + 0.444(-0.2)^n + 5.82(0.8)^n] u[n]}_{\text{zero-state response}} + \underbrace{0.2(-0.2)^n + 0.8(0.8)^n}_{\text{zero-input response}}$$

$$y[n] = \underbrace{-1.26(4)^{-n}}_{\text{forced response}} + \underbrace{0.644(-0.2)^n + 6.62(0.8)^n}_{\text{natural response}}$$



5.7

System Stability

System Stability

External Stability (*BIBO*)

If the input $x[n]$ is bounded, then the output will be bounded if

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Internal Stability (Asymptotic)

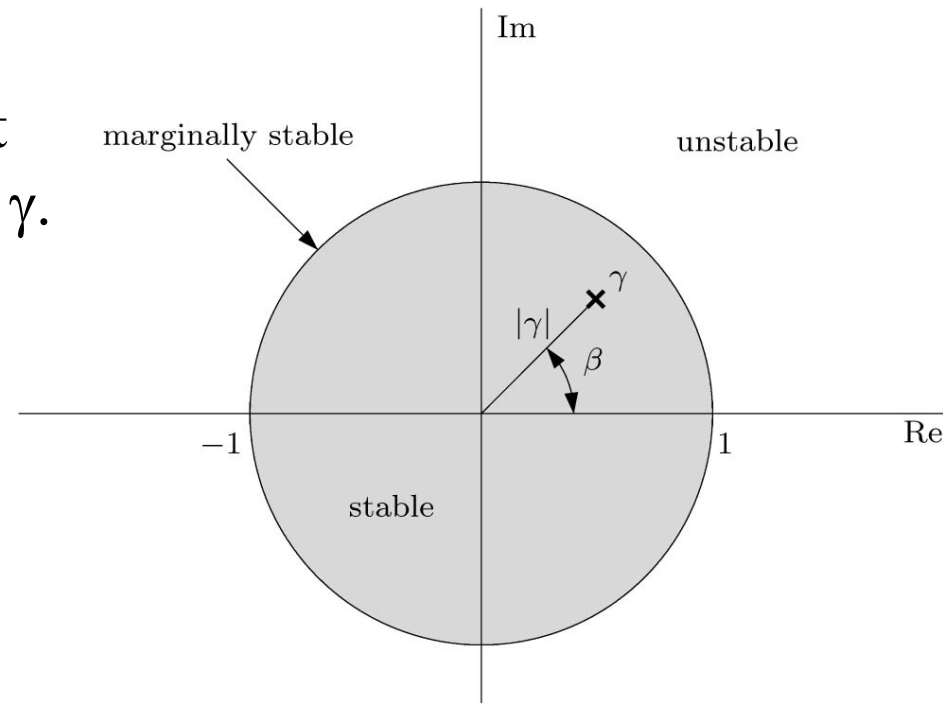
The output of a system consist of a wave similar to the input and a combination of the characteristic modes of the system γ .

$$\gamma = |\gamma|e^{j\beta} \text{ and } \gamma^n = |\gamma|^n e^{j\beta n}$$

if $|\gamma| < 1$, then $\gamma^n \rightarrow 0$ as $n \rightarrow \infty$; **Stable**

if $|\gamma| > 1$, then $\gamma^n \rightarrow \infty$ as $n \rightarrow \infty$; **Unstable**

and if $|\gamma| = 1$, then $|\gamma|^n = 1$ for all n **Marginally Stable**

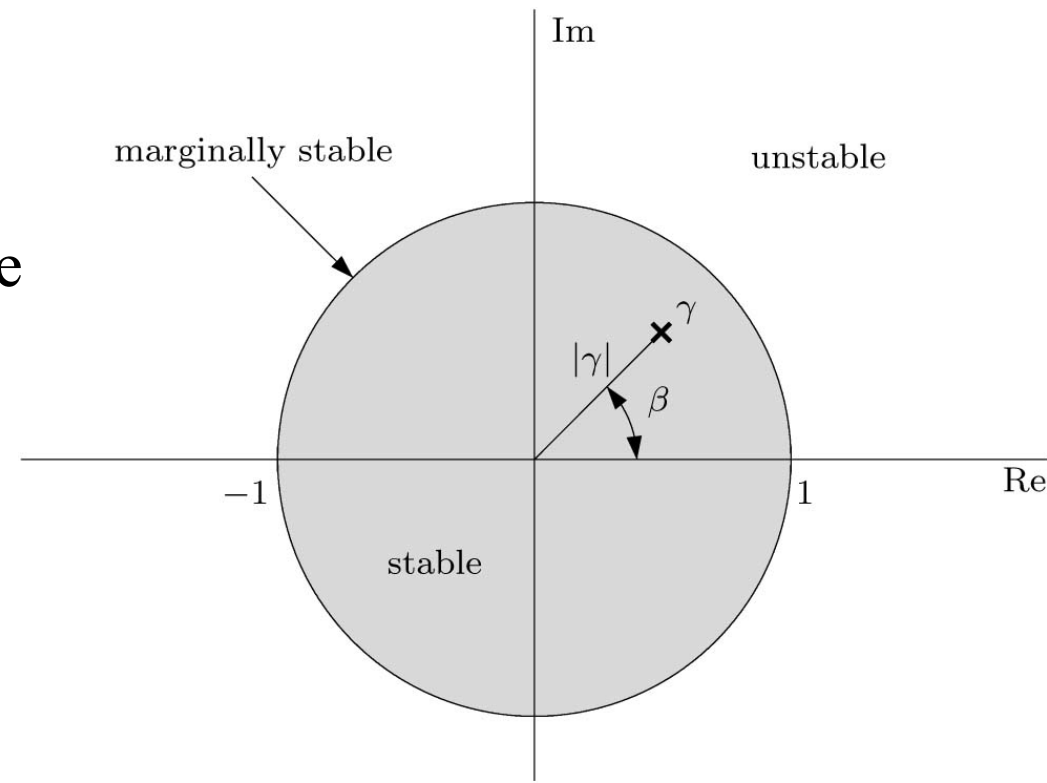


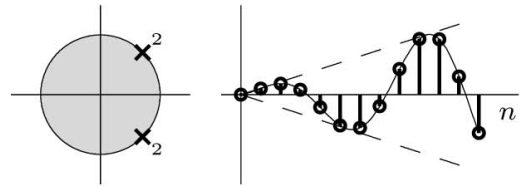
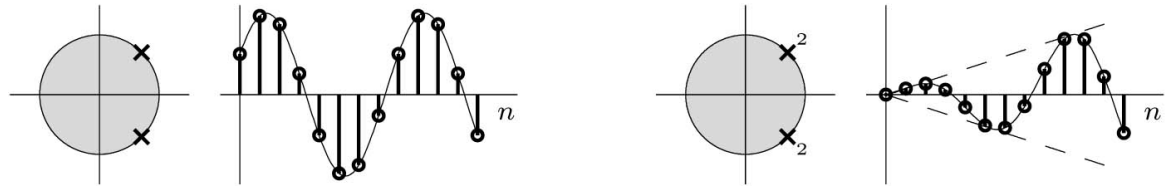
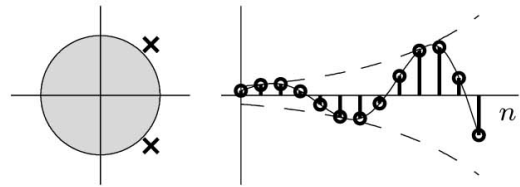
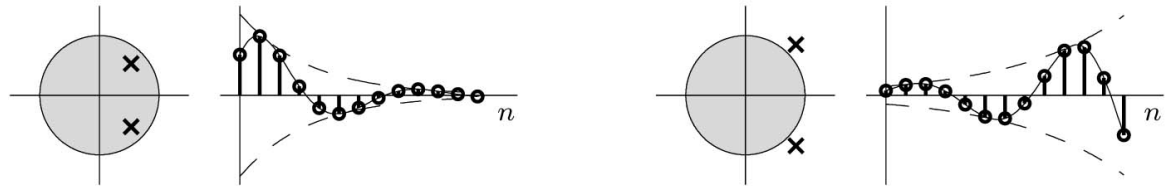
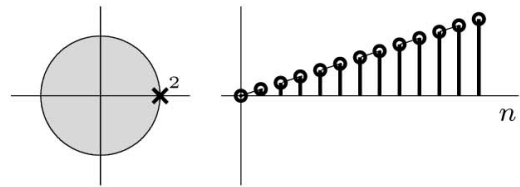
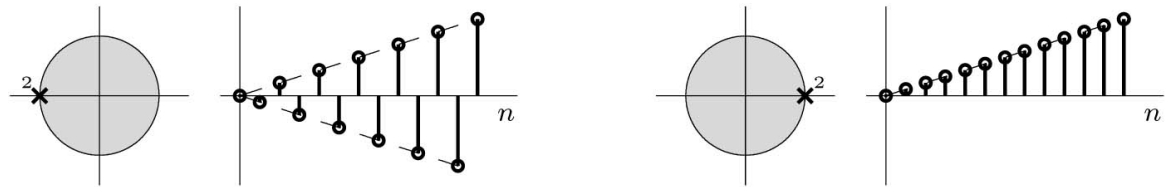
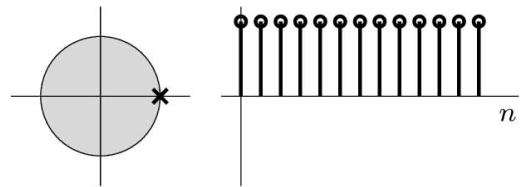
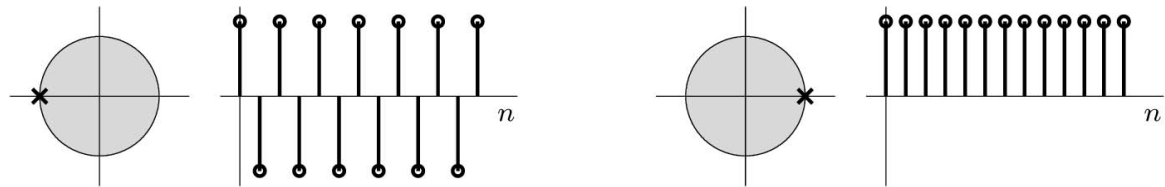
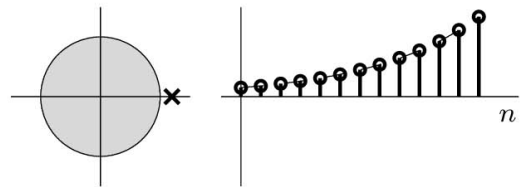
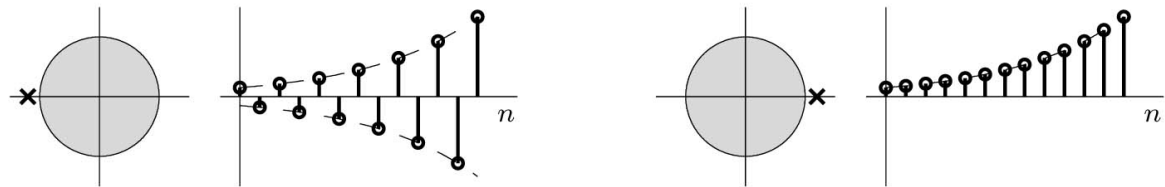
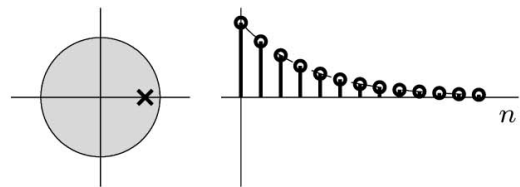
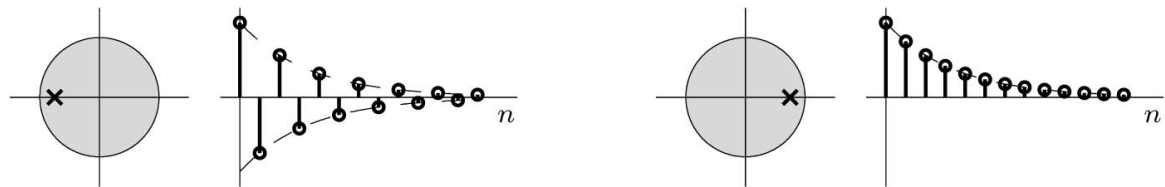
System Stability

1. A causal LTID system is asymptotically stable if and only if all the characteristic roots are inside the unit circle. The roots may be simple or repeated.
2. A causal LTID system is marginally stable if and only if there are no roots outside the unit circle, and there are non-repeated roots on the unit circle.
3. A causal LTID system is unstable if and only if at least one root is outside the unit circle or there are repeated roots on the unit circle or both.

Repeated roots (r roots) on the unit circle will cause the system to be unstable because the characteristic modes will be $n^{r-1}\gamma^n$.

$$n^{r-1}\gamma^n \rightarrow \infty \text{ as } n \rightarrow \infty, \text{ because } |\gamma|^n = 1$$





Example

Determine the internal and external stabilities of the systems specified by the following equations. In each case, plot the characteristic roots in the complex plane. Each system is both controllable and observable.

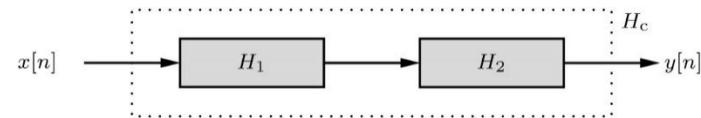
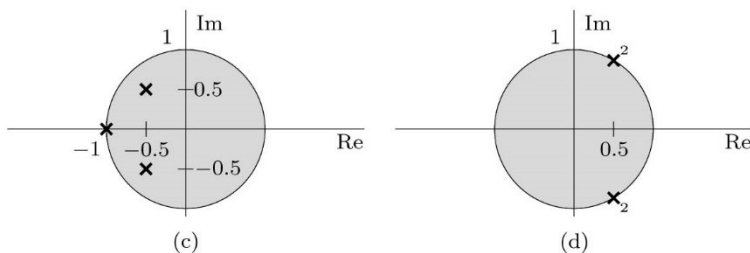
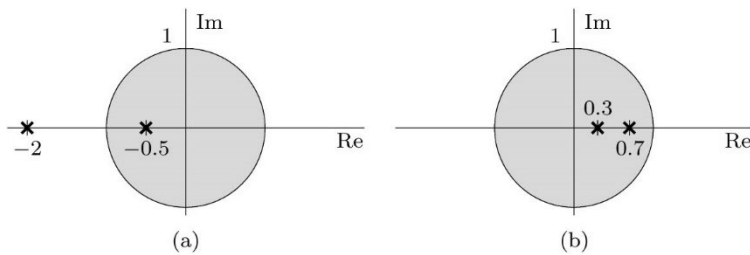
(a) $y[n + 2] + 2.5y[n + 1] + y[n] = x[n + 1] - 2x[n]$

(b) $y[n] - y[n - 1] + 0.21y[n - 2] = 2x[n - 1] + 3x[n - 2]$

(c) $y[n + 3] + 2y[n + 2] + 3/2y[n + 1] + 1/2y[n] = x[n + 1]$

(d) $(E^2 - E + 1)^2 \{y[n]\} = (3E + 1) \{x[n]\}$

(e) Two cascaded systems with $h_1[n] = 4\delta[n] - 3(0.5)^n u[n]$ and $h_2[n] = 2^n u[n]$.



$$h_c[n] = (0.5)^n u[n]$$

5.8

Intuitive Insights into System Behavior

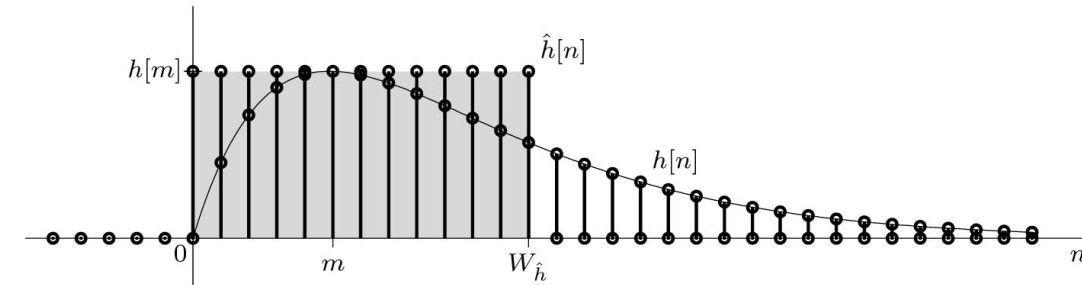
Characteristic Roots and System Behavior

- Dependence of System Behavior on Characteristic Modes.
- Response Time of a System: The System Time Constant.

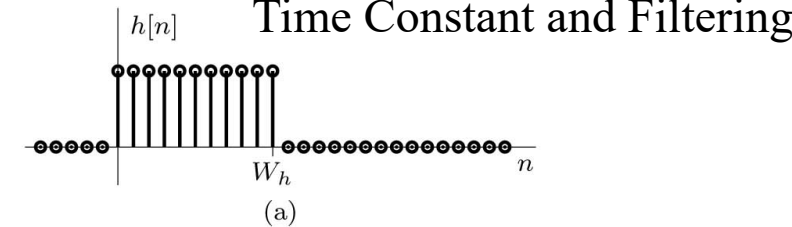
$$(W_{\hat{h}} + 1)h[m] = \sum_{n=-\infty}^{\infty} h[n]$$

- Time Constant and Rise Time of a System.
- Time Constant and Filtering ($W_h \propto 1/f_c$)
- Time Constant and Pulse Dispersion
- The Resonance Phenomenon

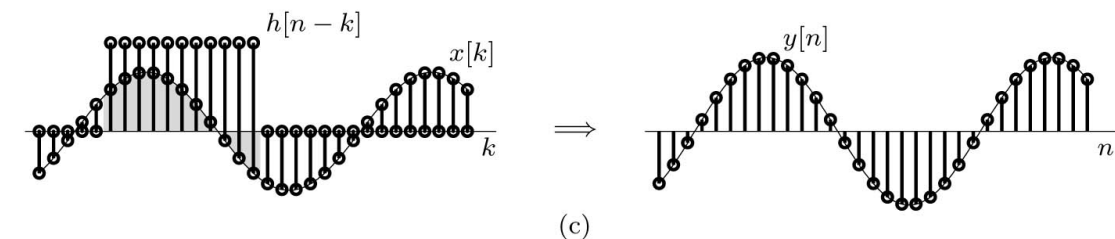
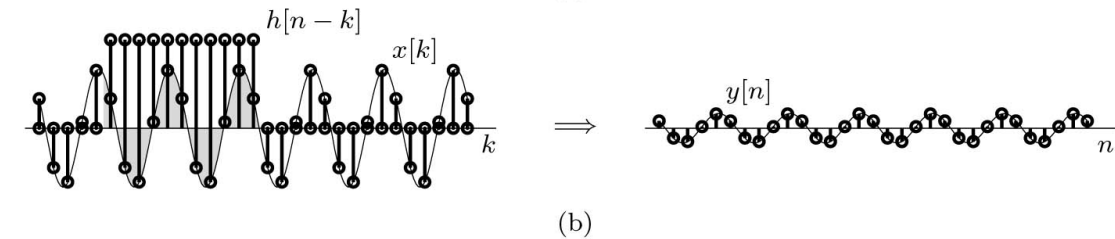
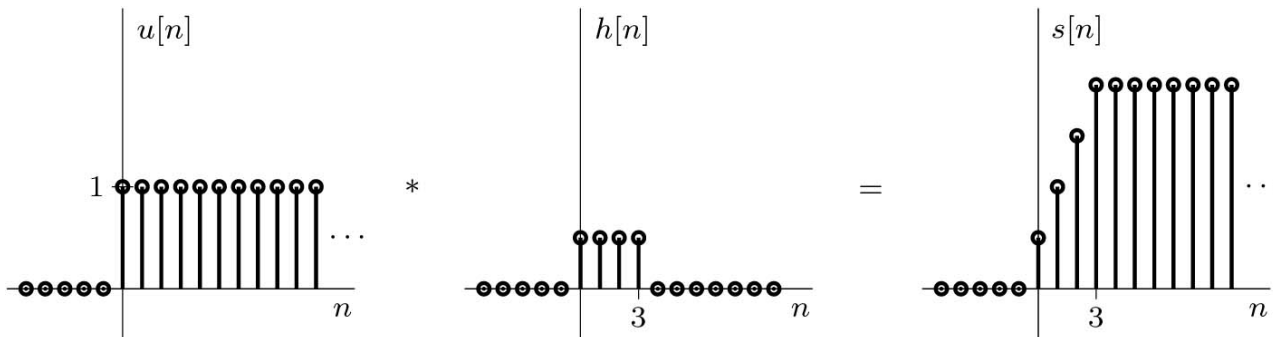
Time Constant Approximation



Time Constant and Filtering



Time Constant as Rise Time



Resonance Phenomenon

The system respond strongly (large output) to an input that is similar or the same as one of the system characteristic modes.

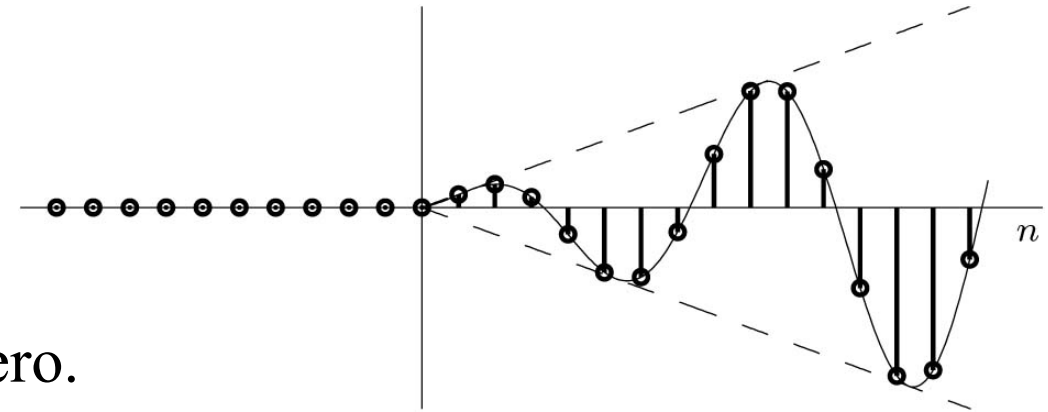
$$h[n] = c\gamma^n u[n]$$

$$x[n] = (\gamma - \epsilon)^n u[n]$$

$$y[n] = c\gamma^n u[n] * (\gamma - \epsilon)^n u[n]$$

$$y[n] = \frac{c [\gamma^{n+1} - (\gamma - \epsilon)^{n+1}]}{\epsilon} u[n]$$

$$\lim_{\epsilon \rightarrow 0} y[n] = c(n+1)\gamma^n u[n]$$



If $|\gamma| < 1$ then the output $y[n]$ will eventually goes to zero.

If $|\gamma| = 1$ then the output will oscillate and grow to infinity.